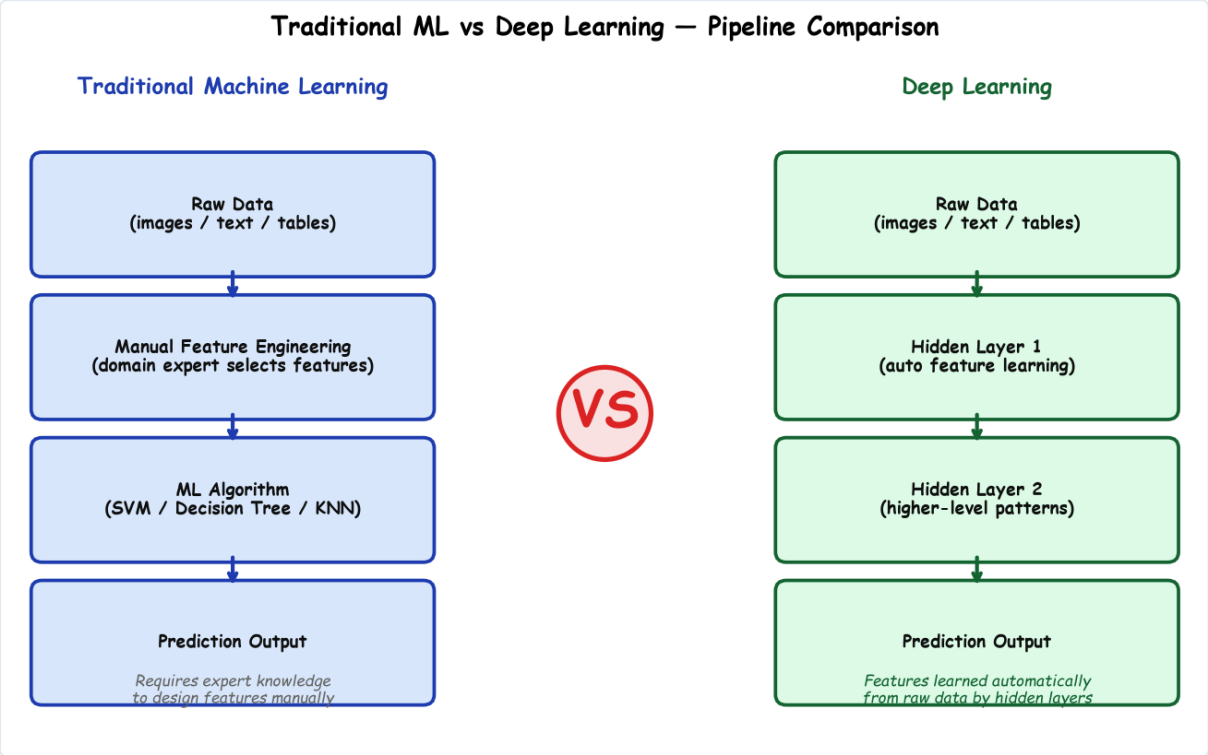
Submodule 36.02: Machine Learning vs Deep Learning

1. Fundamental Conceptual Differences

While both Machine Learning and Deep Learning learn patterns from data, their architectural workflows, data dependencies, and feature extraction mechanisms are fundamentally distinct.



---

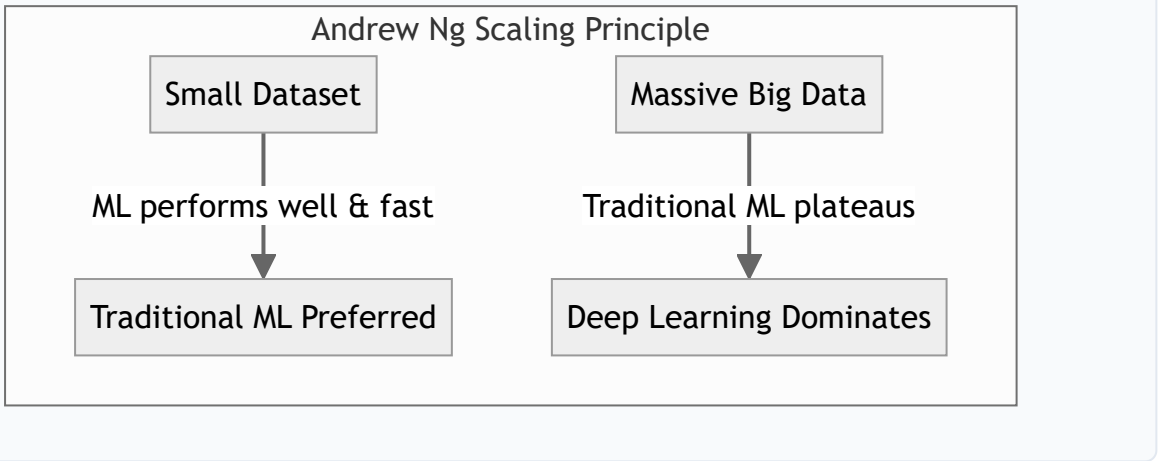
2. The Core Differentiator: Feature Engineering

- **Traditional Machine Learning**: Requires human engineers to manually craft numerical features (e.g., calculating pixel histograms, extracting SIFT/HOG descriptors for images, or computing TF-IDF vectors for text). If the engineered features are poor, the ML model performs poorly regardless of its complexity.
- **Deep Learning (End-to-End Representation Learning)**: Eliminates manual feature engineering. Early neural network layers automatically detect primitive low-level features (e.g., edges, gradients), intermediate layers compose them into mid-level shapes (e.g., textures, contours), and deeper layers assemble them into high-level semantic concepts (e.g., faces, wheels, objects).

---

3. Performance vs Amount of Data (Andrew Ng's Curve)

A defining characteristic distinguishing ML and DL is how model performance scales with dataset size:



- **Traditional ML Algorithms** (e.g., SVM, Random Forest, Logistic Regression): Performance improves initially with data, but quickly **plateaus**. Adding millions of additional records yields diminishing returns.
- **Deep Neural Networks**: Performance continues to scale almost linearly with increasing data volume and network capacity.

---

4. Detailed Dimension-by-Dimension Comparison

Dimension	Traditional Machine Learning	Deep Learning
Dataset Size Required	Performs effectively on small to medium datasets (thousands of rows)	Requires massive datasets (millions of samples) to avoid severe overfitting
Feature Extraction	**Manual Feature Engineering** required by human domain experts	**Automated Feature Learning** performed hierarchically by hidden layers
Hardware Dependency	Runs efficiently on standard multi-core **CPUs**	Requires high-performance **GPUs / TPUs** with high VRAM bandwidth
Training Duration	Fast (seconds to minutes)	Prolonged (hours, days, or weeks depending on parameter count)

Dimension	Traditional Machine Learning	Deep Learning
Model Interpretability	High (**Glass Box** models; feature importances, decision paths clear)	Low (**Black Box** models; millions of non-linear weights make reasoning opaque)
Data Modality	Best for structured, tabular data (Excel sheets, SQL tables)	Dominates unstructured data (high-resolution images, raw audio, video, text)